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DELIVERABLE 3 – Selected problem

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PROBLEM DEFINITION

The selected problem focuses on the unpredictable episodes of Freezing of Gait (FoG) and subsequent fall risks in patients with Parkinson's Disease. FoG is characterized by a sudden, episodic inability to produce effective stepping. The underlying root causes of this problem include:

- Lack of automaticity in the motor process
- Environmental overstimulation
- Sudden direction changes
- Struggles with gait initiation
- Emotional triggers such as anxiety or agitation

Currently, patients lack autonomous, real-time technological interventions to break these paralyzing episodes.

RELEVANCE

Magnitude of the problem

Parkinson's disease affects approximately 11 million people globally (0.15 - 0.2% of the world population). Our primary target audience comprises the estimated 10% of patients with early-onset or juvenile Parkinson's (under 50 years old), who strongly desire to maintain an active, independent lifestyle without relying on full-time caregivers.

Costs and consequences of inaction

Leaving this problem unresolved leads to severe physical injuries from traumatic falls, resulting in high hospitalization risks. Furthermore, it generates significant financial and emotional burdens due to the loss of independence, requiring expensive full-time caregivers, family assistance, or nursing home (RSA) placement. Current treatments are primarily pharmaceutical, which can cause adverse drug interactions (e.g., with diabetes medications) and do not provide real-time mechanical relief.

Impact

Solving this problem offers a technological alternative to purely pharmaceutical or caregiver-dependent approaches. It significantly enhances patient autonomy and shifts the long-term financial costs from expensive continuous assistance to accessible technology.

Uniqueness of the problem

FoG is highly unpredictable, triggered by daily environmental or emotional factors. Its uniqueness lies in the fact that once a patient enters a freezing state, they cannot consciously decide to "unfreeze" or overcome the paralysis on their own without external sensory stimuli.

Innovation potential

There is a strong opportunity to leverage and integrate existing hardware (such as ankle sensors and bone-conduction earphones) into a novel system. This system can accurately detect specific movements, recognize the FoG phenomenon, and instantly deliver personalized cueing techniques to restore movement.

FEASIBILITY ANALYSIS

Specificity

The problem is highly specific, addressing the exact biomechanical and neurological triggers of FoG (overstimulation, direction changes, anxiety) with a targeted intervention system, providing an innovative alternative to existing solutions.

Team characteristics

Developing a trained machine learning system supported by biotechnological expertise perfectly aligns with the team's internal, multidisciplinary competencies.

Existing technology

The necessary technologies are already available. This includes position and physiosomatic (cardiac) sensors for patient state acquisition, as well as smartphones, audio devices, ankle bracelets for muscular impulses, and visual lasers for user interaction and cueing.

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Regulatory landscape

The primary legal challenge involves GDPR compliance, as the system processes sensitive biometric and health data. Strict privacy protocols must be implemented for diagnosing and inserting patient data into databases to improve machine learning training.

Required resources

The team has realistic access to the fundamental resources needed, particularly the ability to train the initial machine learning algorithms on existing datasets of Parkinson's patients.